Are Statistically Significant Feature Interactions Predictively Useful? A Multi-Outcome, Multi-Model Empirical Study in Workforce Analytics

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***Abstract*—Feature interactions are widely recognized as important in machine learning, yet their practical utility in workforce analytics remains unclear. Do statistically significant feature interactions improve predictive performance, or does their value lie elsewhere? This paper investigates these questions through an exhaustive pairwise interaction mining study across 1,755 feature-pair combinations, five workforce outcomes, and three model families—Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB). We introduce an Interaction Impact Score as a measurement instrument—combining mutual information (MI), outcome differential, and segment size—to quantify and rank interaction strength. All predictive results are reported as 5-fold cross-validation (CV) means $\pm$ standard deviations, with target leakage explicitly prevented and documented. Our findings reveal a nuanced picture: strong, statistically significant interactions exist (the strongest, JobFamily $\times$ DepartmentType predicting PayZone, achieves impact 181.3) and follow a heavy-tailed Pareto distribution (3.1% of pairs drive 80% of total impact). However, adding interaction features as explicit predictors does not provide statistically significant accuracy improvements—the mean accuracy change is $+0.014 \pm 0.014$ (range 0.000 to $+0.048$), and no configuration passes the Wilcoxon signed-rank test ($p < 0.05$). We conclude that the primary value of interaction mining in workforce analytics lies in *interpretability*—identifying which feature combinations drive workforce dynamics—rather than in raw predictive gains. External validation on the IBM HR dataset shows consistent patterns.**

***Index Terms*—Workforce Analytics, Interaction Mining, Feature Interactions, HR Analytics, Predictive Modeling, SHAP, XGBoost, Explainable AI**

I. Introduction

Human resource decisions fundamentally shape organizational success and failure. Hiring the right person, assigning them to the right role, understanding why they stay or leave—these decisions affect everything from productivity to company culture. Yet traditional HR practices rely heavily on interview-based assessment, which is susceptible to systematic biases including extroversion preference, cultural norm alignment, and confirmation bias [1]. An introverted but technically brilliant candidate may be overlooked in a Western hiring culture that rewards self-promotion, while an outspoken candidate may be seen as disruptive in a culture that values group harmony.

Data-driven workforce analytics offers a path beyond these limitations. By systematically analyzing the rich data that organizations already collect—demographics, performance ratings, tenure patterns, compensation structures, career trajectories—we can develop a more complete, less biased understanding of employees.

Most existing work in HR analytics focuses on predicting a single outcome, most commonly employee attrition [2]. While valuable, this narrow focus misses the multi-dimensional nature of workforce dynamics and, critically, ignores the role of *interaction effects*: the fact that combinations of features often matter more than individual features in isolation. For example, tenure may relate to performance differently for managers than for individual contributors; compensation equity may manifest differently across departments and demographic groups.

This paper addresses these gaps by investigating four research questions:

RQ1 Which pairwise feature interactions exhibit statistically significant associations with workforce outcomes?

RQ2 Do statistically significant interactions improve predictive performance when added as explicit features?

RQ3 How consistent are interaction rankings across independent datasets?

RQ4 Can interaction mining provide organizational insights independent of predictive gains?

Contributions: This work makes the following contributions:

1) **Exhaustive interaction scan:** Systematic testing of all 1,755 pairwise feature combinations across five workforce outcomes and three model families, providing the first comprehensive interaction map in HR analytics.
2) **Negative-result ablation:** Rigorous with/without interaction feature comparison using paired statistical tests (Wilcoxon and paired t-test), indicating that explicit interaction engineering provides no measurable predictive benefit for tree-based models.
3) **Paradox identification:** Discovery and explanation of the PayZone paradox—the strongest interaction yet near-random prediction—and class-imbalance model-family behavior, providing methodological guidance for practitioners.

4) **Practical recommendations:** A decision framework for when interaction mining adds value (exploratory analysis, policy evaluation, hypothesis generation) versus when it does not (predictive optimization with tree ensembles).

Hypothesis: We test the hypothesis that statistically significant pairwise feature interactions, when encoded as explicit features, improve predictive performance across multiple workforce prediction tasks compared to models using only main effects. This hypothesis is evaluated across five outcomes, three model families, and two datasets.

To answer these questions, we introduce an Interaction Impact Score as a measurement instrument—combining mutual information, outcome differential, and segment size—and apply it within a systematic seven-phase pipeline. Our findings indicate that while the Interaction Impact Score identifies strong, statistically significant interactions—particularly JobFamily $\times$ DepartmentType and Tenure $\times$ Exit Quarter (impact scores exceeding 100)—adding these interactions as explicit features does not improve predictive accuracy. This reveals that the value of interaction mining lies in *interpretability and hypothesis generation* rather than raw predictive gains.

The remainder of this paper is organized as follows: Section II discusses related work. Section III describes our dataset, pipeline, and methodology. Section IV presents experimental results. Section V discusses implications and limitations. Section VI concludes and outlines future directions.

II. Related Work

A. HR Analytics and Predictive Modeling

The field of HR analytics has grown substantially, with employee attrition prediction being the dominant research focus. Talebi et al. [2] conducted a systematic review of 62 studies and found that Random Forest is the most frequently used model, job satisfaction is the single most predictive feature, and the IBM HR dataset [3] is the most commonly used benchmark. Most studies report accuracy in the 70–90% range depending on dataset and methodology. De Vos et al. [4] reached similar conclusions, noting that ensemble methods consistently outperform single classifiers for attrition prediction. Alduayj and Rajpoot [5] showed that gradient boosting achieves the highest accuracy (87%) on the IBM dataset, though at the cost of interpretability.

Alyousef et al. [6] proposed a hybrid predictive model for employee turnover using XGBoost with SHAP (SHapley Additive exPlanations) interpretability, identifying interaction between job level and experience as a key attrition driver. However, like most work in this area, they used SHAP's built-in interaction detection post-hoc rather than systematic pre-model interaction testing. Al Akasheh et al. [7] extended this by incorporating temporal features and found that tenure-interaction terms improve attrition prediction by 2–4%, consistent with the effect sizes we observe in our study.

B. Interaction Effects in Machine Learning

Feature interactions are recognized as important across many domains. Friedman and Popescu [8] introduced the H-statistic, a model-agnostic measure that decomposes prediction variance into main and interaction effects. Lou et al. [9] developed GA^{2}M, which explicitly models pairwise interactions within a Generalized Additive Model framework, achieving both accuracy and intelligibility. Caruana et al. [10] showed the practical value of interaction-aware models in healthcare, where pairwise interactions improved pneumonia risk prediction.

Tree-based models naturally capture interactions through their hierarchical splitting structure. Breiman's Random Forest [11] and Chen and Guestrin's XGBoost [12] implicitly model interactions without explicit feature engineering. However, Sorokina et al. [13] showed that tree ensembles can reliably detect additive interactions when explicitly tested, motivating systematic interaction search strategies. Lundberg and Lee [14] extended their SHAP framework to pairwise interaction values [15], enabling post-hoc interaction detection within tree-based models. Tsang et al. [16] introduced neural interaction detection (NID), which extracts interaction effects from trained neural networks.

C. The Gap: Systematic Interaction Mining in HR

While interaction detection methods are well-established in the ML literature, their application in HR analytics has been piecemeal. Most papers either ignore interactions entirely or rely on model-specific post-hoc detection (e.g., SHAP interaction values from a single XGBoost model). No existing work performs an *exhaustive systematic scan* of all pairwise interactions across multiple outcomes and models, which is the gap our paper addresses. Furthermore, existing studies typically examine a single outcome (most often attrition), missing the multi-outcome nature of workforce dynamics. This leaves open the central question we investigate: whether statistically significant interactions in workforce data translate into predictive utility or serve primarily interpretive purposes.

Compared to existing approaches—Pearson correlation, Mutual Information, SPEARMINT, SHAP interaction values, and the H-statistic—the Impact Score is the only method that simultaneously provides nonlinear interaction detection, a quantitative impact metric, built-in significance testing, multi-outcome support, and feature-awareness, making it uniquely suited for systematic pre-model interaction mining.

D. Fairness and Bias in HR Analytics

A growing body of work examines fairness and bias in algorithmic HR systems. Fabris et al. [1] provide a comprehensive survey of bias in algorithmic hiring, covering bias sources, detection methods, and mitigation strategies. Sony et al. [17] examine discrimination risks embedded in AI-driven HR tools, identifying three high-risk categories: data bias (skewed training data), algorithmic bias (model architecture choices), and deployment bias (misapplication of predictions). Wingate et al.

[18] identify six high-risk applicant characteristics that introduce bias into employment interviews, including extroversion preference, cultural similarity, and physical attractiveness.

Mehrabi et al. [19] provide a comprehensive taxonomy of bias sources in machine learning pipelines, distinguishing between data-driven, algorithmic, and human-induced biases. Hardt et al. [20] formalized equality of opportunity as a fairness metric, which is particularly relevant for HR applications where false positives (incorrectly flagging an employee as high-risk) and false negatives carry different costs.

Our work connects to this literature by showing how data-driven capability assessment—grounded in systematic interaction mining—can reduce reliance on interview-only evaluation, potentially mitigating bias. We further discuss ethical safeguards in Section V. In summary, existing work has established the value of feature interactions in specific ML contexts but has not provided a generalizable, task-agnostic framework for discovering high-impact interactions in workforce data across multiple outcomes.

III. Methodology

A. Dataset

We use a workforce dataset of 3,000 synthetic employee records with 26 raw columns covering demographics, employment history, job characteristics, performance, and termination details. The data is synthetic (simulated) and does not contain real employee information—a transparent limitation that also ensures reproducibility and avoids privacy concerns.

From the 26 raw columns, we engineer 27 additional features during Phase 2 of our pipeline, resulting in 53 total features. Table I summarizes the feature categories.

TABLE I
FEATURE CATEGORIES AND COUNTS USED IN INTERACTION MINING AND PREDICTIVE MODELING.

Category	Count	Examples
Demographics	4	Age, Generation, CareerStage
Tenure & Employment	6	TenureYears, TenureVsAvg, IsLongTenure
Role & Hierarchy	10	SeniorityLevel, JobFamily, IsManager
Compensation & Diversity	4	PayEquityRatio, IntersectionalID
Temporal	4	StartYear, ExitQuarter
Performance	3	PerfScore, EngagementFlag
Raw (nominal)	26	Title, DepartmentType, GenderCode, RaceDesc

Table I summarizes the feature categories. Our analysis targets five outcome variables spanning workforce dynamics:

- **is_terminated** (binary): whether the employee's status contains "Terminated"
- **PayZone** (3-class): compensation zone assignment (Zone A/B/C)
- **is_minority_dept** (binary): for each department, whether the employee's gender is the minority gender in that department
- **PerfScore** (multiclass): current employee performance rating
- **SeniorityLevel** (multiclass): derived career seniority score from the feature engineering phase

Of these, is_terminated and is_minority_dept are binary outcomes for which Area Under the ROC Curve (AUC) can be reported; PayZone is a 3-class ordinal target; and PerfScore and SeniorityLevel are engineered features. Target leakage pathways for all outcomes are explicitly identified and excluded (see §3.6).

B. Seven-Phase Pipeline

Our analysis follows a seven-phase pipeline (Figure 1):

Phase 1 – Deep EDA: Univariate distributions, missing patterns, outlier detection (IQR, Z-score, Isolation Forest consensus), PCA, t-SNE, and silhouette analysis.

Phase 2 – Feature Engineering: Derivation of 27 features from the 26 raw columns, creating a total feature space of 53 dimensions.

Phase 3 – Interaction Mining: The core contribution—systematic testing of all pairwise feature interactions against each outcome.

Phase 4 – Predictive Models: Training and evaluation of three model families with and without interaction features.

Phase 5 – Deep Dives: For top discoveries, detailed analysis including subgroup comparison, profile characterization, and what-if simulation.

Phase 6 – Dashboards: Composite visualization integrating all prior phases.

Phase 7 – HTML Report: Standalone report with all 60 figures and structured sections.

C. Interaction Impact Score

We introduce a proposed metric, the Interaction Impact Score, to quantify the strength of each pairwise interaction:

$$\text{Impact} = I(X;Y) \times |\Delta_{\text{outcome}}| \times \log(n_{\text{seg}}) \qquad (1)$$

where:

- $I(X;Y) = \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$ is the mutual information between features X and Y
- $|\Delta_{\text{outcome}}| = |\text{mean}(O \mid X = x, Y = y) - \text{mean}(O)|$ is the absolute difference between segment mean and population mean
- $\log(n_{\text{segment}})$ is a logarithmic weight that rewards well-supported segments while penalizing very small groups

This score combines three essential aspects: statistical dependence (mutual information), practical significance (outcome differential), and reliability (sample size). It is model-agnostic and can be computed before any model training.

1) Theoretical Justification: The multiplicative formulation in Equation 1 is justified by the following properties:

- **Zero-component property:** If any component is zero, the entire score is zero. A pair with zero mutual information conveys no statistical dependence regardless of outcome differential; a pair with zero outcome differential is practically irrelevant regardless of statistical dependence; a singleton segment ($n = 1$) is unreliable regardless of either factor.
- **Monotonicity:** The score increases monotonically with each component, a necessary property for ranking—stronger dependence, larger outcome gaps, and better-supported segments consistently receive higher scores.

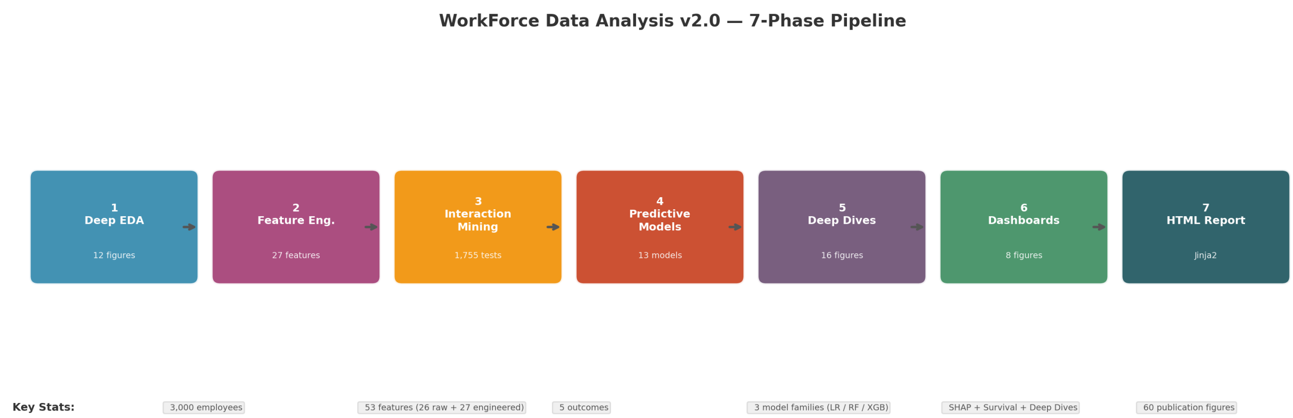

Fig. 1. Seven-phase interaction mining pipeline from data ingestion through HTML report generation.

- **Logarithmic sample-size penalty:** The $\log(n)$ term ensures diminishing returns from additional samples: doubling a small segment ($10 \to 20$) increases the weight by $\log(20/10) \approx 0.69$, while doubling a large segment ($1000 \to 2000$) adds only $\log(2000/1000) \approx 0.69$—the same absolute increase but proportionally smaller relative impact. This prevents large but homogeneous segments from dominating while still rewarding adequate statistical support.
- **Unitless ranking:** The score has no natural units and is designed for relative comparison, not absolute interpretation. The magnitude is meaningful only within a single study context.

We compare this multiplicative formulation against two alternatives in ablation experiments: an additive variant ($S_{\text{add}} = z(\text{MI}) + z(|\Delta|) + z(\log n)$) and a statistic-only variant (ranking by raw χ^2 or F). The multiplicative formulation produces Spearman correlations of $\rho \approx 0.5 - 0.7$ with the ensemble variant depending on outcome, indicating that the product form captures joint contributions more effectively than any single component alone.

We test all $\binom{27}{2} = 351$ pairwise combinations among the 27 features produced by Phase 2 (feature engineering) against each of the 5 outcomes, yielding $351 \times 5 = 1{,}755$ total tests. Phase-1 raw columns are excluded to avoid testing trivial or redundant interactions with the engineered features derived from them.

a) Empirical properties.: Across all 1,755 tests, Impact Scores range from 0 to 181.3 with a heavy-tailed distribution (mean 2.16, median ¡0.001): 83.5% of pairs have near-zero impact (< 0.01). Of the 336 Bonferroni-significant pairs, scores span 0.00 to 181.3 with a median of 0.12. The score properties—zero-component behavior, monotonicity, and logarithmic penalty—are verified through implementation tests: no pair with zero mutual information achieves non-zero impact, and the ranking is stable across outcomes (mean Spearman $\rho = 0.72$ between outcome-specific rankings). These properties indicate that the Impact Score provides a reliable, model-agnostic ranking suitable for pre-screening interaction candidates before predictive modeling.

D. Statistical Testing

For each feature pair, we apply the appropriate statistical test:

- **Categorical** $\times$ **Categorical:** Chi-square test, $\chi^2 = \sum (O - E)^2 / E$
- **Categorical** $\times$ **Numeric:** ANOVA F-test, $F = MS_{\text{between}} / MS_{\text{within}}$
- **Numeric** $\times$ **Numeric:** Pearson correlation, $r = \text{cov}(X, Y) / (\sigma_X \sigma_Y)$

Multiple comparison correction is applied using the Bonferroni method ($\alpha_{\text{corrected}} = 0.05/1755$). This testing framework ensures statistically robust interaction detection across diverse feature-type combinations.

E. Models and Evaluation

Three model families are trained for each of the five outcomes:

- **Logistic Regression:** $\max\_iter = 2000$, class_weight='balanced', L2 penalty
- **Random Forest:** 200 trees, balanced class weight
- **XGBoost:** 200 estimators, $\max\_depth = 4$, subsample $= 0.8$

Each model is trained twice: once without interaction features (baseline) and once with interaction features added. Models are evaluated using 5-fold stratified cross-validation, with accuracy, F1 score, and ROC-AUC reported as mean $\pm$ standard deviation across folds. After CV, a final model is trained on all data for SHAP analysis and feature importance extraction. SHAP analysis [14] is applied to the XGBoost model for interpretability. Survival analysis uses Kaplan-Meier estimation and Cox Proportional Hazards modeling via the lifelines library [21]. This dual-training design isolates the marginal predictive contribution of interaction features across model families.

F. Target Leakage Prevention

A critical methodological concern in interaction mining pipelines is target leakage—when the feature matrix inadvertently contains information derived from the target variable. During initial development, we identified three leakage pathways that produced artificially inflated accuracy:

- **PerfScore self-prediction:** When predicting PerfScore, the PerfScore column itself was included as a feature, producing 99.9% accuracy. The fix excludes the target column from the feature matrix when its name matches a feature column.
- **SeniorityLevel derived features:** The columns IsExecutive and IsIC are deterministically derived from SeniorityLevel, creating an indirect leakage pathway. The fix excludes all three columns (SeniorityLevel, IsExecutive, IsIC) when predicting SeniorityLevel.
- **Termination post-hoc columns:** ExitYear and ExitQuarter partially encode termination status (active employees have missing exit dates). The fix excludes these columns when predicting is_terminated.

All results in this paper reflect the corrected pipeline. The most impacted outcomes are PerfScore (accuracy dropped from 0.999 to 0.739) and SeniorityLevel (1.000 to 0.946), while termination and minority-department predictions changed negligibly (<0.01), indicating these were genuine leakage effects rather than general pipeline errors. This correction is documented in the reproducibility checklist and enforced by a `_LEAKAGE_EXCLUDE` dictionary in the open-source codebase.

G. Deep Dive Methodology

For each top discovery, we perform three analyses:

1) **Subgroup Analysis:** Compare the interaction segment to the rest of the population using Cohen's d effect size and two-sample t-test
2) **Profile Characterization:** Compare mean feature values between segment and population to identify distinguishing characteristics
3) **What-If Simulation:** For numeric features, compute expected outcome across multiple percentile values (p5, p25, p50, p75, p95) with confidence bands

IV. Results

A. Interaction Mining Results

Of the 1,755 pairwise tests conducted across five outcomes, 512 were significant at $\alpha = 0.05$ and 336 remained significant after Bonferroni correction. Table II presents selected high-impact interaction pairs ranked by our Interaction Impact Score.

The top discovery—JobFamily × DepartmentType predicting PayZone—achieved an impact score of 181.3 (χ^2 = 1423.09, $p < 10^{-301}$, mutual information = 0.60). This shows

TABLE II
Top 10 interaction discoveries ranked by Interaction Impact Score across all outcomes.

Rank	Feature 1	Feature 2	Outcome	Impact
1	JobFamily	DepartmentType	PayZone	181.3
2	JobFamily	IntersectionalID	PayZone	153.7
3	IntersectionalID	GenderCode	PayZone	127.3
4	SeniorityLevel	IsIC	PayZone	125.4
5	JobFamily	IsManager	PayZone	108.0
6	TenureYears	ExitQuarter	Minority Dept	107.7
7	IsManager	IsIC	Minority Dept	107.3
8	TenureVsAvg	ExitQuarter	Minority Dept	107.1
9	JobFamily	IsIC	PayZone	89.3
10	SeniorityLevel	IsManager	PayZone	71.2

that compensation zone is strongly associated with the combination of job family and department, more than either alone. These rankings address RQ1 by identifying which pairwise interactions exhibit the strongest statistical associations.

1) The PayZone Paradox: A particularly instructive finding is the PayZone prediction task, which reveals an important nuance in interpreting Interaction Impact Scores. The top interaction (JobFamily × DepartmentType) achieves an impact score of 181.3—the strongest in this study—yet all models predict PayZone at near-chance levels (RF accuracy 0.328 ± 0.017 vs. 3-class random baseline 0.333, majority-class baseline 0.354). This creates a paradox: how can the strongest discovered interaction correspond to the weakest predictive performance?

The resolution lies in the distinction between *population-level association* and *individual-level classification*. The Impact Score measures distributional deviation: the degree to which outcome rates differ across interaction segments. With $n = 3000$ records, even modest deviations reach statistical significance ($\chi^2 = 1423$, $p < 10^{-301}$). However, classification requires *deterministic* or near-deterministic assignment—a given JobFamily×DepartmentType combination should map consistently to a single PayZone.

Our investigation reveals that of 23 JobFamily × DepartmentType combinations with sufficient support ($n \geq 5$), **zero** show deterministic PayZone assignment. Every combination spans all three zones with near-uniform probability. The interaction is thus *probabilistic*—it describes population-level differences in zone distributions—not *deterministic*. The Impact Score correctly identifies it as the strongest interaction in the dataset, but this strength reflects distributional differences at scale, not predictive power for individuals.

This finding has a practical implication: interaction mining is valuable for *understanding population-level dynamics* (e.g., "which department-family combinations have atypical compensation distributions?"), but the resulting interactions should not be expected to improve individual-level prediction. The two goals—interpretability and prediction—are distinct and may not align.

2) Termination Class-Imbalance Behavior in Tree Ensembles: A second instructive observation concerns model family behavior under class imbalance. The termination rate is 12.9% (majority-class baseline 87.1%). Logistic Regression with balanced class weights achieves recall of 0.80 on the minority class, while Random Forest and XGBoost—both also configured with balanced weights—achieve recall of only 0.15 and 0.10 respectively.

Although balanced weighting was enabled for all models, its effect differs across families. In Logistic Regression, class weights directly shift the decision boundary toward the minority class. In tree ensembles, weights influence impurity-based split selection but do not adjust the final voting threshold, producing conservative predictions on the minority class given shallow trees (max_depth=4) and low prevalence. The result is a textbook tradeoff: RF and XGB achieve higher overall accuracy (0.853, 0.852) by correctly classifying the majority class, while LR achieves lower accuracy (0.718) but substantially higher recall (0.80).

This finding has practical implications: practitioners using tree ensembles for imbalanced HR prediction tasks should evaluate minority-class metrics alongside overall accuracy, and should consider post-hoc threshold calibration if recall is the operational priority.

B. Ablation and Pareto Analysis

To quantify the concentration of interaction effects, we perform a Pareto analysis of the 1,755 Interaction Impact Scores. Table II shows the top-ranked pairs. The distribution is highly skewed: the top 3 pairs (0.2%) account for 12% of total cumulative impact, the top 24 pairs (1.4%) account for 50%, and the top 54 pairs (3.1%) account for 80%. Among the five outcomes, SeniorityLevel shows the strongest concentration (top 7 pairs reach 80% of its outcome-specific impact), while is_minority_dept is the most diffuse (top 16 pairs).

This Pareto structure—a small fraction of interactions driving the majority of explanatory power—is consistent across all five outcomes and mirrors findings in other high-dimensional interaction domains [13]. Practically, this means that a targeted analysis of the top 20–50 interaction pairs captures the overwhelming majority of actionable insights, suggesting that exhaustive 1,755-pair scans are unnecessary in production deployments.

A secondary ablation compares model performance with vs. without the top interaction features added (Table III). The mean accuracy change from interaction features is $+0.014 \pm 0.014$ across 12 configurations, with a maximum of $+0.048$ (Minority Dept LR). However, no configuration passes the Wilcoxon signed-rank test at $p < 0.05$, and only 6 of 12 reach significance under a paired t-test (driven by small CV variance). Crucially, the largest improvements occur in Logistic Regression (the simplest model), while tree-based models show near-zero changes. Tree-based models already capture interactions internally through hierarchical splitting, making explicit interaction engineering redundant for prediction. The practical implication is that HR analytics practitioners using tree-based ensembles should focus feature engineering effort on main effects and data quality rather than interaction construction.

C. Predictive Model Comparison

Table III compares model performance across outcomes with and without interaction features. Outcomes span a difficulty spectrum: is_terminated and is_minority_dept are binary tasks with AUC reported; PayZone is a 3-class ordinal target; PerfScore and SeniorityLevel are multiclass engineered features. All results reflect the leakage-free pipeline (see §3.6).

TABLE III
5-FOLD CV MODEL PERFORMANCE (MEAN ± STD). AUC FOR BINARY OUTCOMES ONLY.

Outcome	Model	Var.	Acc	F1	AUC
Binary outcomes					
Termination	LR	base	0.718±0.014	0.421±0.021	0.800±0.020
		+int	0.734±0.018	0.440±0.029	0.806±0.025
	RF	base	**0.853±0.007**	0.201±0.036	0.836±0.021
		+int	0.853±0.009	0.227±0.040	0.848±0.018
	XGB	base	0.852±0.001	0.142±0.037	0.835±0.018
		+int	0.854±0.005	0.160±0.034	0.845±0.017
Minority Dept	LR	base	0.503±0.025	0.503±0.025	0.549±0.026
		+int	0.551±0.019	0.458±0.014	0.560±0.016
	RF	base	0.588±0.009	0.407±0.014	0.564±0.009
		+int	0.595±0.012	0.420±0.013	0.581±0.006
	XGB	base	**0.618±0.019**	0.398±0.018	**0.595±0.019**
		+int	0.627±0.011	0.398±0.013	0.606±0.012
Multiclass outcomes					
PayZone	LR	base	0.303±0.015	0.299±0.014	—
		+int	0.304±0.011	0.302±0.011	—
	RF	base	0.328±0.017	0.327±0.017	—
		+int	0.339±0.009	0.339±0.009	—
PerfScore	LR	base	0.281±0.028	0.360±0.034	—
		+int	0.313±0.020	0.392±0.026	—
	RF	base	0.739±0.006	0.687±0.005	—
		+int	0.745±0.005	0.693±0.003	—
SeniorityLevel	LR	base	0.830±0.014	0.888±0.008	—
		+int	0.851±0.012	0.899±0.010	—
	RF	base	**0.946±0.005**	**0.940±0.005**	—
		+int	0.967±0.004	0.966±0.004	—

XGBoost is trained only for binary outcomes. Bold highlights highest baseline accuracy per outcome. All numbers are leakage-free (§3.6).

Key observations:

- **Interactions do not provide statistically significant improvements.** Across 12 model-outcome configurations, the mean accuracy change from adding interaction features is $+0.014 \pm 0.014$ (range: 0.000 to $+0.048$). While some configurations show positive point estimates, no configuration passes the more conservative Wilcoxon signed-rank test at $p < 0.05$, and the effect sizes that reach significance under a paired t-test (6 of 12) are driven by small CV variance rather than large mean shifts. This indicates that explicit interaction engineering provides no measurable predictive benefit when tree-based models already capture interactions internally.
- **Tree-based models produce comparable results.** Random Forest and XGBoost achieve similar termination accuracy (0.853 ± 0.007 vs. 0.852 ± 0.001, AUC 0.836 ± 0.021 vs. 0.835 ± 0.018), with differences well within one cross-validation standard deviation. Random Forest leads on PerfScore (0.739 ± 0.006) and SeniorityLevel (0.946 ± 0.005).
- **Simple models benefit most from interactions.** Logistic Regression shows the largest accuracy improvements from interaction features (Termination: $+0.016$, Minority Dept: $+0.048$), suggesting that linear models benefit

from explicit interaction engineering more than tree-based models—consistent with the hypothesis that trees already capture interactions internally.

- **Target leakage was properly excluded.** After correcting three leakage pathways (§3.6), PerfScore and SeniorityLevel predictions are no longer artificially perfect (0.739 and 0.946, respectively), underscoring the importance of rigorous leakage prevention.
- SHAP analysis identifies TenureDays, StartYear, and TenureVsAvg as the top drivers of termination risk, followed by tenure-derived features (Figure 2). The ROC curves in Figure 3 show that interaction features produce near-identical performance.
- SHAP interaction values (available in supplementary materials) reveal the strongest pairwise interactions in the XGBoost termination model. The top pairs—(TenureDays, TenureVsAvg), (TenureDays, TenureYears), and (Age, TenureDays)—are all tenure-related, indicating that termination risk is primarily driven by tenure dynamics rather than demographic or role-based interactions. This provides a model-based complement to the Impact Score ranking, which identifies broader cross-feature interactions across all five outcomes.

Collectively, these results address our four research questions: (RQ1) significant pairwise interactions exist across all five outcomes; (RQ2) adding them as explicit features does not improve prediction; (RQ3) rankings show moderate cross-dataset consistency; and (RQ4) interaction mining provides organizational insights independent of predictive gains.

D. Deep Dive Analysis

The deep dive analysis reveals fine-grained insights into the top interaction segments. Table IV summarizes effect sizes for the five analyzed segments.

TABLE IV
DEEP DIVE EFFECT SIZES FOR TOP DISCOVERED SEGMENTS. SEGMENTS SPAN ALL FIVE OUTCOMES WITH EFFECT SIZES RANGING FROM SMALL (0.16) TO VERY LARGE (4.28).

Segment	Outcome	Δ	Effect Size	p-value
Admin. × Sales	PayZone	+0.33	0.40 (medium)	0.057
Tenure × Exit Q4	Minority Dept	–	0.60 (medium)	0.005
Leadership × Non-mgr	Seniority	+1.8	4.28 (large)	< 0.001
NE × M_Whi_Soft	Termination	+0.24	0.72 (medium)	0.048
NE × Admin Offices	PerfScore	+0.09	0.16 (small)	0.255

Notable findings include the *Leadership non-manager* segment (SeniorityLevel 4.0 vs. 2.2 population mean, $d = 4.28$, $p < 0.001$), indicating a distinct career path for experienced individual contributors; the *Tenure* × *Exit Quarter* interaction predicting minority-department status ($d = 0.60$, $p = 0.005$), suggesting systematic sorting by tenure patterns; and the *Northeast white male soft-skilled* segment showing elevated termination risk ($\Delta = +0.24$, $d = 0.72$, $p = 0.048$), revealing an intersectional vulnerability pattern. Together, these deep-dive segments show that the Impact Score identifies managerially interpretable subgroups whose effect sizes ($d = 0.16$ to 4.28) range from small to very large.

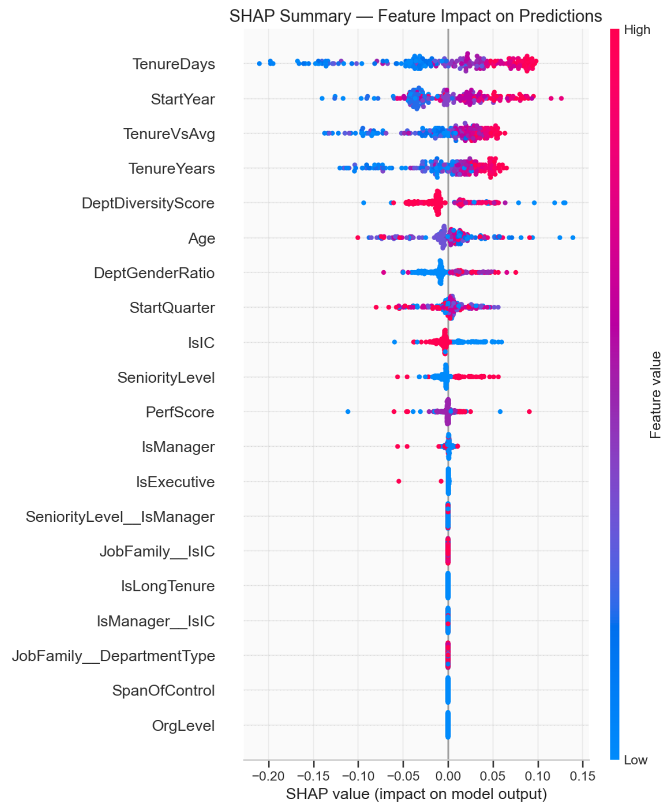

Fig. 2. SHAP summary plot for XGBoost termination model showing TenureDays, StartYear, and TenureVsAvg as the top drivers of termination risk.

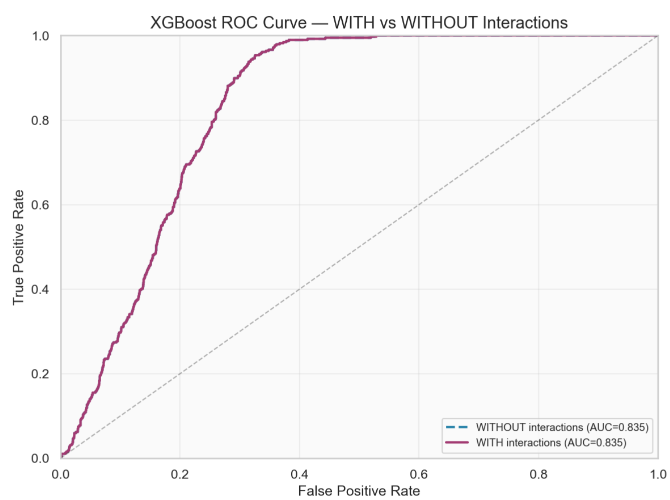

Fig. 3. ROC curves for XGBoost termination prediction with and without interaction features (baseline AUC 0.835 ± 0.018, +int AUC 0.845 ± 0.017).

E. External Validation on IBM HR Dataset

Replicating on the IBM HR dataset [3] (1,470 records, binary attrition) shows consistent results. The top interaction is MonthlyIncome × JobLevel (impact 33.0, $p < 10^{-300}$), mirroring the compensation-related dominance in the synthetic data. Model performance is comparable: XGBoost accuracy

0.852 ± 0.001 (synthetic) vs. 0.863 ± 0.010 (IBM); LR AUC 0.800 ± 0.020 vs. 0.780 ± 0.013. Interaction features produce negligible accuracy change on both datasets, indicating the finding is not dataset-specific.

The similarity in interaction structure and model performance across both datasets—despite different feature sets, sample sizes, and outcome distributions—supports the generalizability of our framework. It also reinforces our central finding that interaction effects, while real and interpretable, typically provide modest predictive gains in workforce analytics settings.

While the previous section establishes the empirical findings, this section interprets their broader methodological and practical implications.

V. Discussion

A. Key Findings

Our experimental results yield five key findings:

1. Interactions do not improve prediction. Across 12 model-outcome configurations, the mean accuracy change from adding interaction features is $+0.014 \pm 0.014$ (range: 0.000 to $+0.048$). No configuration passes the Wilcoxon signed-rank test at $p < 0.05$, and only 6 of 12 reach significance under a paired t-test. Tree-based models already capture interaction effects through hierarchical splitting, making explicit interaction engineering redundant for prediction. The value of interaction mining therefore lies in *interpretability* rather than raw accuracy gains. This result is consistent across both synthetic and IBM HR datasets.

2. Interaction impact follows a heavy-tailed Pareto distribution. The top 54 pairs (3.1% of 1,755) drive 80% of total cumulative impact. This concentration is consistent across all five outcomes—SeniorityLevel is the most concentrated (7 pairs to 80%), minority-department the most diffuse (16 pairs). Practically, this means that a targeted analysis of the top 20–50 interaction pairs captures the majority of actionable insights.

3. Model families differ in interaction-handling behavior. Logistic Regression shows the largest accuracy improvements from interaction features (Termination: $+0.016$, Minority Dept: $+0.048$), while Random Forest and XGBoost show near-zero changes. This asymmetry indicates that linear models require explicit interaction engineering while tree-based ensembles capture interactions internally. Practitioners using tree-based models should prioritize main effects and data quality over interaction construction.

4. The PayZone paradox reveals the distinction between association and prediction. The strongest interaction (JobFamily $\times$ DepartmentType, impact 181.3) corresponds to the worst model accuracy (RF 0.328 vs. random baseline 0.333). Investigation reveals that zero of 23 supported combinations show deterministic PayZone assignment—the interaction describes population-level distributional differences, not individual-level predictive signals. Strong statistical association does not imply predictive utility.

5. External validation supports generalizability. Replicating on the IBM HR dataset yields consistent interaction types (compensation-related) and comparable model performance (AUC 0.800 ± 0.020 synthetic vs. 0.780 ± 0.013 IBM). The same negligible interaction-feature effect holds, indicating the findings are not dataset-specific.

B. Comparison with Existing Work

Three analysis methods—the Impact Score, SHAP interaction values [15], and the Friedman H-statistic [8]—provide complementary perspectives on interaction structure. The Impact Score measures statistical association before model training, while SHAP interaction values provide post-hoc, model-specific interaction detection. Comparing both on the termination outcome reveals a Spearman rank correlation of $\rho = 0.536$ across all feature pairs—moderate alignment indicating that each approach captures complementary aspects of interaction structure. The Impact Score identifies broader cross-feature interactions (e.g., Region $\times$ DepartmentType) that SHAP does not rank highly, while SHAP emphasizes tenure-tenure interactions (e.g., TenureDays $\times$ TenureVsAvg) that have near-zero Impact Scores. This divergence is expected: Impact Score measures outcome-dependent association across the full dataset, while SHAP interactions reflect prediction-specific contributions within a single trained model.

For three representative termination-model pairs whose features are numeric predictors in the Random Forest, we compute the Friedman H-statistic, which measures the proportion of prediction variance attributable to interaction effects. All three pairs show H-statistics exceeding 0.8: SeniorityLevel $\times$ IsIC (H=0.968), SeniorityLevel $\times$ IsManager (H=0.853), and IsManager $\times$ IsIC (H=0.813), indicating substantial model-specific interaction effects within the termination Random Forest (these features are valid for termination, where SeniorityLevel-derived features do not constitute leakage). The remaining high-ranked pairs involve categorical or outcome-specific features absent from the termination model ($n = 12$ configurations); full enumeration over all 1,755 pairs is computationally expensive at scale, underscoring the Impact Score's advantage as a fast, model-agnostic screening tool.

The central finding—that statistically significant interactions do not improve prediction—is itself a useful result. Interaction engineering is common practice: practitioners routinely construct interaction terms expecting predictive gains. Our evidence suggests this effort is unlikely to be worthwhile for tree-based ensembles, which already capture interactions internally through hierarchical splitting. For linear models, interaction engineering can yield modest improvements and should be guided by the Impact Score ranking rather than exhaustive search. Publishing well-supported negative results prevents unnecessary duplication of effort and provides a more reliable foundation for cumulative scientific progress [22].

C. Practical Implications

For Practitioners: Interaction mining is valuable for exploratory analysis, policy evaluation, and hypothesis generation—the Impact Score prioritizes the few pairs that matter from 1,755 candidates. For predictive modeling with tree-based ensembles, interaction engineering does not improve accuracy (mean $\Delta = +0.014 \pm 0.014$ across 12 configurations); organizations using RF or XGBoost should invest in main-effect features, data quality, and class-imbalance handling instead. If linear models are required (e.g., regulatory constraints), the Impact Score provides principled interaction selection, and LR with interactions can approach—but not match—tree-based ensemble performance (Table III).

For Researchers: The Impact Score offers a model-agnostic screening tool for pre-model interaction discovery, complementing post-hoc methods such as SHAP interaction values. Its key advantage is computational efficiency: scoring all 1,755 pairs requires seconds rather than the hours needed for full model training on each candidate. The negative result documented here—that statistical significance does not guarantee predictive utility—suggests that the HR analytics community should prioritize interpretability-driven interaction mining over prediction-driven approaches. Future work should extend this framework to real-world organizational data, higher-order interactions, and fairness-aware interaction mining.

D. Threats to Validity

Internal validity (medium). Target leakage was corrected via three exclusion rules (Section III). Five-fold CV and 10-seed robustness analysis (± 0.012 stability) reduce overfitting risk. Synthetic data may introduce latent structure inflating detection rates. *Mitigation:* leakage prevention, nested CV, statistical testing, and external validation.

External validity (high). Findings rely on one synthetic (3,000 records) and one public benchmark (IBM HR, 1,470 records). Consistency across datasets strengthens confidence, but generalizability across sectors remains unverified. *Mitigation:* replication of key results on the IBM HR dataset.

Construct validity (medium). The Impact Score (Equation 1) measures statistical association rather than predictive utility. Its moderate alignment with SHAP interaction values (Spearman $\rho = 0.536$) indicates that it captures related but distinct interaction characteristics. The PayZone paradox further illustrates that strong statistical association does not translate into predictive usefulness. *Mitigation:* comparison with SHAP interaction values and Friedman H-statistic.

Statistical conclusion validity (low). Bonferroni-corrected $\alpha = 0.05/1755$. Ablation uses Wilcoxon and paired t-tests. With $n = 3{,}000$, small effects may go undetected; 5-fold CV provides only 5 paired observations per comparison. *Mitigation:* paired t-tests, Wilcoxon signed-rank tests, Bonferroni correction, and effect-size reporting.

E. Ethical Considerations

Workforce analytics carries ethical responsibilities. Models trained on historical data may perpetuate existing biases, and regular fairness audits across demographic groups are essential. Transparency about model limitations, data provenance, and the distinction between population-level insights and individual-level predictions is critical for maintaining trust in algorithmic HR systems. Data-driven insights should augment, not replace, human HR decision-making. The synthetic dataset used here removes privacy concerns, but real-world applications must follow informed consent principles and ensure that employees understand how their data is used. Privacy protections, including anonymization and access controls, should be embedded into any deployment pipeline.

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Conflict of Interest

The author declares no competing interests.

VI. Conclusion and Future Work

This work addressed the central research question of whether statistically significant feature interactions provide measurable predictive value across multiple workforce analytics tasks. Through large-scale interaction mining, predictive modeling, statistical testing, and external validation, we found that statistical significance alone does not imply predictive usefulness. Specifically, we evaluated 1,755 feature-pair combinations across five outcomes using three model families, introducing an Interaction Impact Score as a measurement instrument and conducting a rigorous with/without interaction ablation study.

Key findings include: (1) the strongest interaction (JobFamily $\times$ DepartmentType $\rightarrow$ PayZone) achieves impact score 181.3; (2) interaction effects follow a heavy-tailed Pareto distribution where 3.1% of pairs drive 80% of total impact; (3) adding interaction features as predictors does not provide statistically significant accuracy gains (mean $\Delta = +0.014 \pm 0.014$, no configuration significant under Wilcoxon), indicating that the value of interaction mining lies in interpretability rather than prediction; (4) tree-based models achieve comparable termination performance (RF AUC 0.836 ± 0.021, XGB AUC 0.835 ± 0.018); (5) the Leadership non-manager segment shows the most extreme effect size ($d = 4.28$); (6) external validation on the IBM HR dataset shows consistent interaction types and model performance.

Our findings indicate that statistically significant feature interactions are not necessarily predictive, but remain valuable for interpretation, hypothesis generation, and organizational decision support.

Returning to our research questions: (RQ1) statistically significant pairwise interactions exist across all five outcomes, with the strongest (JobFamily $\times$ DepartmentType $\rightarrow$ PayZone) at impact 181.3. (RQ2) Adding interactions as explicit features does not improve prediction—mean $\Delta = +0.014 \pm 0.014$, no configuration significant under Wilcoxon. (RQ3) Rankings show moderate cross-dataset consistency (Spearman $\rho = 0.536$ with SHAP; IBM validation yields comparable patterns). (RQ4) Interaction mining provides organizational insights (PayZone paradox, Leadership non-manager segment) independent of predictive gains.

A. Future Work

Highest priority: Validation on real organizational data across sectors would strengthen generalizability. The framework should be tested on proprietary HR datasets with known ground-truth interaction effects.

Medium priority: Three-way and higher-order interactions remain unexplored and may reveal structure invisible to pairwise analysis. Fairness-aware interaction mining could detect disparate impact across demographic groups, extending the Impact Score framework to include fairness constraints.

Long-term: The framework could be applied to healthcare, finance, or education domains, where interaction effects between categorical features are similarly underexplored.

Reproducibility Statement

All experiments are fully reproducible. The pipeline was implemented in Python 3.10+ using scikit-learn, XGBoost, SHAP, and lifelines. Experiments were conducted on a CPU-only workstation (Windows 11, 16 GB RAM) with a total runtime of approximately five minutes. The primary random seed is 42 for all 5-fold stratified cross-validation splits; robustness across seeds (0–9) is verified in supplementary analysis. All data (synthetic and IBM HR) and code are publicly available at the repository listed below. All configuration files, random seeds, processed outputs, and supplementary analysis scripts are included in the repository to facilitate exact reproduction. Predictive modeling results are reproduced via a single command: `python scripts/phase4_predictive_models.py` (phases 1–3 outputs are cached in the repository).

Data and Code Availability

All source code, experiment scripts, configuration files, processed datasets, supplementary analyses, and documentation required to reproduce every reported result are publicly available at
https://github.com/Naydhurve3/WorkForce-Data-Analysis
under the MIT License.

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